Akanksha Cirigiri

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Professional Summary

Early-career Software Engineer with strong foundation in Java, SQL, and backend development. Experienced in designing and implementing enterprise-grade solutions, debugging and optimizing code, and collaborating in global teams. Skilled in problem-solving, cloud basics, and distributed systems, with hands-on experience in full-stack and AI/ML projects. Committed to learning and applying best engineering practices in high-impact environments.

Education

Jawaharlal Nehru Technological University, Hyderabad — Nov 2021 – Jun 2025
B.Tech in Artificial Intelligence and Machine Learning — **CGPA: 8.2/10**

Technical Skills

Programming: Java, Python, C
Problem Solving: Data Structures & Algorithms, OOPS
Backend Development: Spring Boot, REST APIs, JDBC, SQL
Tools & Platforms: Git, GitHub, Linux (Ubuntu), VS Code, Postman, Docker (Basic), AWS (Basic)
Data & AI: Machine Learning, NLP, Computer Vision, CNNs, Transfer Learning

Professional Experience

Centre for Development of Advanced Computing (C-DAC) — Hyderabad, India
AI & ML Intern — Aug 2025 – Present

- Designed and evaluated **hybrid quantum–classical anomaly detection models** using analytical problem-solving.
- Collaborated with cross-functional teams to analyze problems and benchmark solutions against metrics.
- Executed end-to-end **data processing and validation**, improving model accuracy by **16%**.

EY Global Services — Hybrid
Full Stack Intern — Mar 2025 – Apr 2025

- Developed full-stack components in a **team-based Agile environment**.
- Built backend APIs and responsive UI, improving efficiency by **30%**.
- Ranked in the **Top 10%** of interns for performance and collaboration.

Projects

Blood Cell Classification – Deep Learning Pipeline — Code Link
Tech Stack: Python, TensorFlow, OpenCV, Transfer Learning

- Designed and implemented an end-to-end **machine learning pipeline** involving data preprocessing, exploratory analysis, model training, validation, and evaluation for automated blood cell classification.
- Applied **transfer learning techniques** to enhance model performance, improving classification accuracy by **15%** over baseline models.
- Focused on **quality metrics, validation strategies, and performance reporting** to ensure reliable and reproducible results.

Breast Ultrasound Anomaly Detection — Code Link
Tech Stack: Python, OpenCV, CNN, Isolation Forest, Random Forest

- Analyzed medical ultrasound image data and engineered **morphological and texture-based features** to improve anomaly detection accuracy from **26% to 95%**.
- Trained and evaluated **classical machine learning models and CNN architectures**, ensuring robust performance assessment.
- Assessed model effectiveness using **confusion matrix, precision, recall, and F1-score** to support data-driven conclusions.

The Maze of the Worm (Java Game) — Code Link
Tech Stack: Java, OOP, Event-Driven Programming

- Developed an **event-driven Java application** demonstrating strong object-oriented design principles, including modular game logic and state management.
- Optimized application logic to handle **high-frequency update cycles** efficiently, ensuring low-latency input handling and smooth runtime performance.

Publications & Achievements

Cirigiri, Akanksha (May 2025). "*AI-Driven Classification from Peripheral Blood Cell Images for Hematological Disorders.*"

International Journal of Innovative Research in Technology (IJIRT), Volume 11, Issue 12, ISSN: 2349-6002. **[View Publication]**

Certifications

Certificate of Appreciation from , **EY Global Services** | Developer Job Simulation by, **Accenture** | Software Engineering Job simulation by,**Goldman Sachs** | AI fundamentals & Machine Learning by,**IBM** | Practical Machine learning, **Johns Hopkins University**